

React MCQ

Chapter 1: Introduction to React

1. What is React primarily used for?
 - a) Backend development
 - b) Database management
 - c) **Building user interfaces**
 - d) Server-side scripting
2. Who is the creator of React?
 - a) Google
 - b) **Facebook**
 - c) Microsoft
 - d) Twitter
3. Which of the following is a key feature of React?
 - a) Server-side rendering
 - b) **Component-based architecture**
 - c) Static typing
 - d) SQL querying
4. What does JSX stand for?
 - a) **JavaScript XML**
 - b) JavaScript Extension
 - c) JavaScript XHTML
 - d) JavaScript Example
5. Which command is used to create a new React app using Create React App?
 - a) **npx create-react-app my-app**
 - b) npm install react my-app
 - c) yarn add react-app my-app
 - d) react-create my-app

Chapter 2: React Components

6. In React, what is a component?

- a) **A function that returns JSX**
- b) A CSS class
- c) An HTML element
- d) A JavaScript variable

7. What is the purpose of props in a React component?

- a) To manage state within the component
- b) **To pass data from parent to child components**
- c) To define component styles
- d) To handle user events

8. How do you define a functional component in React?

- a) `function MyComponent() { return <div>...</div>; }`
- b) `class MyComponent extends React.Component { render() { return <div>...</div>; } }`
- c) `const MyComponent = () => <div>...</div>;`
- d) **Both a and c**

9. How do you use state in a class component?

- a) **`this.state = { key: 'value' };`**
- b) `useState({ key: 'value' });`
- c) `const [state, setState] = useState({ key: 'value' });`
- d) `const state = { key: 'value' };`

10. What method is used to update state in a class component?

- a) **`this.setState()`**
- b) `this.updateState()`
- c) `this.changeState()`
- d) `this.modifyState()`

Chapter 3: Hooks

11. What is a hook in React?

- a) A method for handling events
- b) **A way to manage state and side effects in functional components**
- c) A new HTML element
- d) A CSS preprocessor

12. Which hook is used to add local state to a functional component?

- a) useEffect
- b) **useState**
- c) useContext
- d) useReducer

13. What does the useEffect hook do?

- a) Manages component state
- b) **Runs side effects after rendering**
- c) Handles events
- d) Defines component styles

14. Which hook can be used to manage complex state logic in functional components?

- a) useState
- b) useContext
- c) **useReducer**
- d) useMemo

15. How do you use the useState hook?

- a) **const [state, setState] = useState(initialState);**
- b) const state = useState(initialState);
- c) const setState = useState(initialState);
- d) const [state, setState] = useEffect(initialState);

Chapter 4: Routing

16. Which library is commonly used for routing in React applications?

- a) **React Router**

- b) Redux
- c) Axios
- d) Lodash

17. How do you define a route using React Router?

- a) `<Route path="/home" component={Home} />`
- b) **`<Route path="/home" element={<Home />} />`**
- c) `<Route url="/home" component={Home} />`
- d) `<Route url="/home" element={<Home />} />`

18. What is the purpose of BrowserRouter in React Router?

- a) To manage state
- b) **To handle routing based on the URL**
- c) To style components
- d) To fetch data from APIs

19. Which component is used to navigate programmatically in React Router?

- a) Link
- b) NavLink
- c) **useHistory**
- d) Route

20. What does the Switch component do in React Router?

- a) **It renders only the first child route that matches the current location.**
- b) It applies styles to the routes.
- c) It handles data fetching for routes.
- d) It allows for nested routing.

Chapter 5: State Management

21. Which library is commonly used for state management in React applications?

- a) **Redux**
- b) Axios
- c) React Router
- d) Lodash

22. What is the main purpose of Redux in a React application?
- a) To manage routing
 - b) To handle side effects
 - c) **To manage global state**
 - d) To style components
23. What does dispatch do in Redux?
- a) **Sends an action to the store**
 - b) Updates the component state
 - c) Fetches data from an API
 - d) Applies middleware
24. How do you connect a React component to the Redux store?
- a) **Using connect() from react-redux**
 - b) Using useState() from react
 - c) Using withRouter() from react-router
 - d) Using compose() from redux
25. What does mapStateToProps do in a Redux-connected component?
- a) **Maps state from the Redux store to component props**
 - b) Maps dispatch functions to component props
 - c) Maps API responses to component props
 - d) Maps route parameters to component props

Chapter 6: Context API

26. What is the purpose of React's Context API?
- a) To manage local component state
 - b) **To provide a way to pass data through the component tree without prop drilling**
 - c) To handle routing
 - d) To perform side effects
27. How do you create a Context in React?
- a) **const MyContext = React.createContext();**

- b) `const MyContext = new React.Context();`
- c) `const MyContext = useContext();`
- d) `const MyContext = useState();`

28. What does the `Context.Provider` component do?

- a) **Provides context values to its descendant components**
- b) Consumes context values from parent components
- c) Handles routing
- d) Manages global state

29. How do you consume context values in a functional component?

- a) Using `useContext(MyContext)`
- b) Using `Context.Consumer`
- c) Using `this.context`
- d) **Both a and b**

30. Which of the following is true about context in React?

- a) It is a replacement for state management libraries.
- b) **It is used to avoid prop drilling.**
- c) It can only be used with class components.
- d) It requires manual re-rendering.

Chapter 7: Form Handling

31. What is the purpose of controlled components in React?

- a) **To manage form data using React state**
- b) To handle routing
- c) To fetch data from APIs
- d) To style components

32. How do you handle form submissions in React?

- a) **By using `event.preventDefault()` in the form's `onSubmit` handler**
- b) By using `form.submit()` method
- c) By using `useState()` hook
- d) By using `fetch()` method

33. What does `event.preventDefault()` do in a form submission handler?
- a) **Prevents the default form submission behavior**
 - b) Submits the form to the server
 - c) Validates the form fields
 - d) Resets the form fields
34. How do you access form field values in a controlled component?
- a) **Using `event.target.value` in the `onChange` handler**
 - b) Using `useRef` hook
 - c) Using `document.getElementById`
35. How do you initialize state for form fields in a controlled component?
- a) **`const [value, setValue] = useState("");`**
 - b) `const [value, setValue] = useReducer("");`
 - c) `const [value, setValue] = createState("");`
 - d) `const [value, setValue] = useEffect("");`
36. What is the purpose of the `onChange` event handler in a controlled component?
- a) **To update the state with the latest input value**
 - b) To submit the form data
 - c) To validate form inputs
 - d) To reset the form
37. Which hook is commonly used to handle form validation?
- a) `useState`
 - b) `useEffect`
 - c) **`useReducer`**
 - d) `useMemo`
38. In a controlled component, how do you set the value of an input field?
- a) **By setting the value of the input to the state variable**
 - b) By using `document.getElementById()`
 - c) By directly modifying the input's value property
 - d) By using `setState()`

39. What is the key difference between controlled and uncontrolled components?
- a) **Controlled components use React state to manage form data, while uncontrolled components use the DOM.**
 - b) Uncontrolled components use React state, while controlled components use the DOM.
 - c) Controlled components are always stateless, while uncontrolled components manage state.
 - d) There is no difference; both terms are interchangeable.

Chapter 8: Styling

40. Which approach to styling is considered a best practice in React applications?
- a) Inline styles
 - b) **CSS Modules**
 - c) Global CSS files
 - d) jQuery UI
41. How do you apply styles using CSS Modules in React?
- a) **import styles from './styles.module.css'; and use className={styles.className}**
 - b) import './styles.css'; and use className="className"
 - c) import styles from './styles.css'; and use className={styles.className}
 - d) import styles from './styles.module.css'; and use className="className"
42. What is the purpose of CSS-in-JS libraries like styled-components?
- a) **To allow CSS styles to be written in JavaScript**
 - b) To manage state in React applications
 - c) To handle routing in React applications
 - d) To fetch data from APIs
43. How do you use inline styles in a React component?
- a) **By using the style attribute with an object containing CSS properties**
 - b) By adding a style tag inside the JSX
 - c) By importing a CSS file
 - d) By using className

44. Which of the following is a benefit of using CSS Modules?
- a) **Scopes CSS to the component to prevent class name collisions**
 - b) Increases global CSS file size
 - c) Makes styling easier with external libraries
 - d) Automatically handles CSS animations

Chapter 9: Performance Optimization

45. What is the purpose of React.memo?
- a) **To memoize a functional component to prevent unnecessary re-renders**
 - b) To manage global state
 - c) To handle side effects
 - d) To fetch data from APIs
46. How do you prevent a component from re-rendering in React?
- a) **By using React.memo for functional components and PureComponent for class components**
 - b) By using useEffect with an empty dependency array
 - c) By using useState with a constant value
 - d) By using useReducer instead of useState
47. What does the useCallback hook do?
- a) **Memoizes a callback function to prevent re-creation on every render**
 - b) Caches a function's result
 - c) Handles asynchronous code
 - d) Manages component state
48. How do you optimize performance for lists in React?
- a) **By using key prop for each list item**
 - b) By using useEffect to fetch data
 - c) By avoiding functional components
 - d) By using inline styles
49. What is React.lazy used for?

- a) **For code splitting by dynamically importing components**
- b) For handling form submissions
- c) For managing global state
- d) For styling components

Chapter 10: Error Boundaries

50. What is an Error Boundary in React?

- a) **A component that catches JavaScript errors in its child components**
- b) A method for handling API errors
- c) A way to manage component state
- d) A library for routing

51. Which lifecycle method is used in class components to catch errors in Error Boundaries?

- a) **componentDidCatch**
- b) componentWillUnmount
- c) componentDidUpdate
- d) componentDidMount

52. What does the componentDidCatch method return?

- a) **Error information and error info**
- b) State and props
- c) API response data
- d) Component styles

53. Can Error Boundaries be used with functional components?

- a) **No, only class components can be Error Boundaries**
- b) Yes, using useErrorBoundary hook
- c) Yes, with useErrorBoundary hook from a library
- d) Yes, using the useEffect hook

54. How can you handle errors globally in React applications?

- a) **By using a top-level Error Boundary component**

- b) By setting global error handlers in JavaScript
- c) By using try-catch blocks in all components
- d) By implementing a custom hook

Chapter 11: Testing

55. Which testing library is commonly used for testing React components?

- a) **Jest**
- b) Mocha
- c) Jasmine
- d) QUnit

56. What is the purpose of react-testing-library?

- a) **To test React components by simulating user interactions**
- b) To handle global state
- c) To manage routing
- d) To fetch data from APIs

57. How do you test asynchronous code with Jest?

- a) **By using async/await and expect with resolves or rejects**
- b) By using setTimeout and expect
- c) By using done callback
- d) By using waitFor method

58. What is enzyme used for in React testing?

- a) **For shallow rendering and full DOM rendering of React components**
- b) For handling side effects
- c) For managing global state
- d) For routing in React applications

59. How do you mock API calls in tests?

- a) **By using jest.mock to mock the API module**
- b) By using real API endpoints
- c) By using fetch directly in tests
- d) By using axios without mocking

Chapter 12: Deployment

60. Which of the following is a common platform for deploying React applications?

- a) **Vercel**
- b) MySQL
- c) MongoDB
- d) Redis

61. How do you create a production build of a React application?

- a) **By running npm run build**
- b) By running npm start
- c) By running npm install
- d) By running npm test

62. What is the purpose of environment variables in React applications?

- a) **To configure different settings for development and production environments**
- b) To manage routing
- c) To handle user authentication
- d) To style components

63. How can you deploy a React application to GitHub Pages?

- a) **By using the gh-pages package**
- b) By using npm run deploy
- c) By uploading files manually
- d) By using the deploy command in create-react-app

64. What is a common practice for handling API secrets in a React application?

- a) **Storing secrets in environment variables**
- b) Hardcoding secrets in source code
- c) Committing secrets to version control
- d) Including secrets in client-side JavaScript